Model vs Index — CAGR Comparison

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What period this covers

Item	Value
Test window	2020-07-01 to 2025-01-31
Rebalance frequency	monthly
Rebalance dates	54
Elapsed time	4.50 years
Holding period	entry to the next rebalance's entry (chaining)
Portfolio	equal-weighted top 20, rebalanced each period
CAGR formula	product of (1 + period return) over 54 periods, then raised to 1/4.50

Basis

Price return compared with price return. NSE's historical index endpoint serves price indices only — every total-return (TRI) spelling returns no data. The strategy's dividends are therefore reported separately rather than folded into the headline, because comparing a dividend-inclusive strategy against a dividend-less index would credit the model with roughly the market's yield (about 1–1.5% a year in India) for nothing.

Index returns are measured over the sessions the strategy actually traded — the same next-session entry and the same exit — not between month-ends, so neither side is measuring a different span of market time.

Headline: CAGR by strategy

Strategy	Gross	Net	vs UNIVERSE	vs N50	vs N500	vs NMIDCAP 150	vs NSMALLCAP 250
equal weight universe	41.13%	39.68%	-1.45%	+21.20%	+18.25%	+8.99%	+6.84%
growth only	49.51%	46.05%	+4.92%	+27.57%	+24.61%	+15.35%	+13.20%
model 5	73.85%	63.67%	+22.54%	+45.19%	+42.24%	+32.98%	+30.83%
model 5 gated	72.87%	61.85%	+20.72%	+43.37%	+40.41%	+31.15%	+29.01%
momentum only	56.63%	41.84%	+0.71%	+23.36%	+20.40%	+11.14%	+8.99%
quality only	33.22%	31.45%	-9.68%	+12.97%	+10.02%	+0.76%	-1.39%
random ranking	36.71%	11.66%	-29.47%	-6.82%	-9.77%	-19.03%	-21.18%
simple quality momentum	40.51%	31.48%	-9.65%	+13.00%	+10.04%	+0.78%	-1.37%
value only	92.84%	85.83%	+44.70%	+67.35%	+64.40%	+55.14%	+52.99%

vs UNIVERSE is the column that matters, and it is deliberately placed before the index columns. It compares the strategy against an equal-weighted holding of the very same point-in-time universe it selected from, so it isolates stock selection.

Why the index columns flatter every strategy. The eligible universe returned 41.13% against NIFTY 500's 21.43% — a +19.70 point gap earned purely by equal-weighting a broad, small-cap-heavy universe, before any selection at all. That gap is inside every "vs index" number below. A strategy that beats NIFTY 500 by 20 points while matching the universe has selected nothing.

Index reference

Index	CAGR	Volatility	Sharpe	Max drawdown	Periods
NIFTY 50	18.48%	13.76%	1.31	-12.15%	54
NIFTY 500	21.43%	14.30%	1.44	-12.47%	54
NIFTY MIDCAP 150	30.69%	16.71%	1.70	-13.67%	54

NIFTY SMALLCAP 250	32.84%	19.64%	1.56	-18.67%	54
Eligible universe (equal weight)	41.13%	-	-	-	54

The **eligible universe** row is the benchmark that controls for size and breadth: the same point-in-time universe the strategy chose from, held equally weighted. Beating NIFTY 500 while trailing this means the model captured a small-cap tilt rather than stock selection.

Risk-adjusted comparison

Strategy	Basis	CAGR	Volatility	Sharpe	Max DD	Hit rate
equal weight universe	net	39.68%	21.47%	1.68	-15.24%	69%
growth only	net	46.05%	24.98%	1.66	-23.71%	72%
model 5	net	63.67%	25.87%	2.07	-15.61%	74%
model 5 gated	net	61.85%	25.77%	2.03	-15.61%	70%
momentum only	net	41.84%	30.77%	1.30	-33.59%	67%
quality only	net	31.45%	19.29%	1.53	-24.47%	69%
random ranking	net	11.66%	23.84%	0.58	-40.36%	50%
simple quality momentum	net	31.48%	25.15%	1.23	-27.94%	67%
value only	net	85.83%	29.39%	2.30	-18.08%	74%

Excess return against NIFTY 500

Strategy	Mean excess/period	Tracking error	Information ratio	Periods beaten
equal weight universe	+1.30%	14.87%	1.05	63%
growth only	+1.74%	19.72%	1.06	57%
model 5	+2.74%	22.60%	1.46	61%
model 5 gated	+2.64%	22.58%	1.40	63%
momentum only	+1.62%	24.84%	0.78	59%
quality only	+0.74%	14.10%	0.63	52%
random ranking	-0.56%	18.56%	-0.36	48%
simple quality momentum	+0.85%	19.72%	0.52	57%
value only	+3.92%	24.28%	1.94	65%

Information ratio is excess return per unit of tracking error — how reliably the outperformance was earned, not how large it was. **Periods beaten** is the share of rebalances where the strategy outpaced the index; a high CAGR with a low share was carried by a handful of periods.

Turnover and cost drag

Strategy	One-way turnover	Gross CAGR	Net CAGR	Cost drag
equal weight universe	0.049	41.13%	39.68%	1.45%
growth only	0.118	49.51%	46.05%	3.46%
model 5	0.326	73.85%	63.67%	10.18%
model 5 gated	0.379	72.87%	61.85%	11.03%
momentum only	0.486	56.63%	41.84%	14.79%
quality only	0.082	33.22%	31.45%	1.76%
random ranking	0.978	36.71%	11.66%	25.05%
simple quality momentum	0.391	40.51%	31.48%	9.04%
value only	0.164	92.84%	85.83%	7.01%

Cost drag is annualised, and it is the difference between a ranking that works on paper and one that works in an account. A strategy replacing most of its book every month pays it repeatedly.

Before quoting these numbers

An index is investable and this portfolio is a simulation. The index CAGR is achievable through a low-cost fund; the strategy CAGR assumes every fill happened at the modelled price and size.

The window is what it is. A CAGR over a few years is dominated by the market regime inside it. Check the periods-beaten share and the drawdown before treating a CAGR gap as skill.

Costs are modelled, not measured. Statutory charges are exact and effective-dated; half-spread and market impact are assumptions, which is why gross and net are both shown.

Delisting recovery is an assumption. Re-run with --delisting-strategy zero and last_close to see how much of the gap depends on it.